Republic of the Philippines
CAVITE STATE UNIVERSITY
Imus Campus
Cavite Civic Center, Palico IV, City of Imus, Cavite
www.cvsu-imus.edu.ph

DEPARTMENT OF COMPUTER STUDIES
COSC 95 – PROGRAMMING LANGUAGES
PROJECT

Dental Healthcare Management System (Appointment Scheduling and Billing Information)

A Project Documentation
Submitted to the Faculty
of the Department of Information Technology
Cavite State University - Imus Campus
City of Imus, Cavite

In partial fulfillment
of the requirements in the subject,
COSC 95 – Programming Languages

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Introduction

Healthcare Management Systems (HCMS) are systems that allow users to manage the administrative, financial, and clinical aspects of a specific healthcare facility. Often, these systems are used in order to enhance the security of the facility's users (patients and doctors), as well as to improve patient care through the automation of manual processes like appointment scheduling. In some cases, these systems are narrowed down to the specific needs that a specific facility is in need of, offering modular or scalable versions of a larger system in order to comply with the current needs of the clients.

In the subdivision of Villa de Primarosa, located in Buhay na Tubig, Imus, Cavite, a local dental clinic can be located. This specific clinic is called "LE Dental Clinic", a dentistry clinic of which can be found in Phase 5 of the said subdivision. Though their facilities appear to be of high quality, their appointment scheduling is limited to calls or through walk-in appointments.

To counter these issues, the proponents decided to synthesize a scalable version of a HCMS that targets small clinics like LE Dental Clinic, as well as the specific issues these kinds of clinics suffer from. To be specific, the proponents had decided to develop a HCMS that allows the following features: appointment scheduling, billing information, appointment records, patient registrations, and users and services management.

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Scope and Limitations

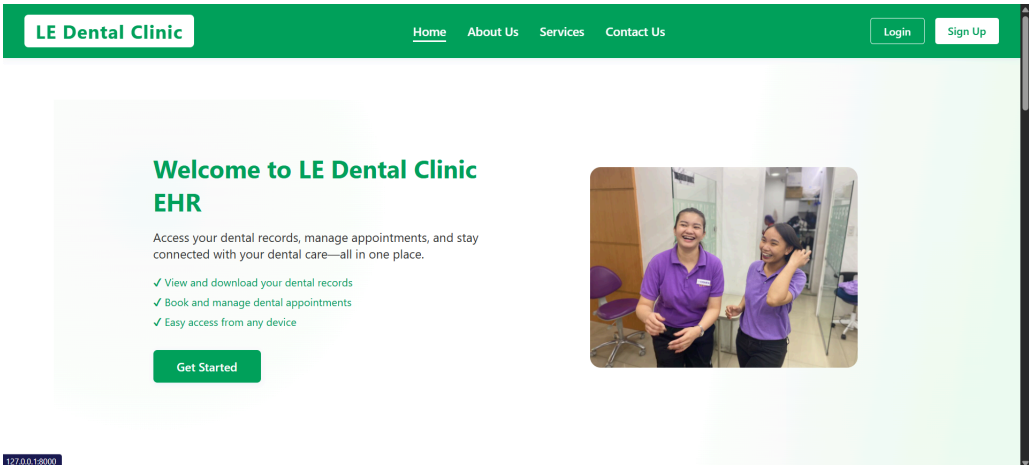
The proposed system's functionality, alluding to the principle of being a scalable version of HCSMS, has been narrowed down to appointment scheduling, billing information, appointment records, patient registrations and services management. If the Patient were to re-schedule their appointments, they would have to cancel their appointment schedule as a whole and remake the appointment on other dates and time. The system will also only be limited to three different user accounts: The Admin, Doctor, and Patient, with each of them having their own set of permissions. The Admin will have the highest level of authority within the system, having access to almost all records and features except appointment scheduling, as patients are the only users that can appoint schedules. As for their permissions, the permissions of both Doctors and Patients are limited to what permissions the Admin will give their accounts.

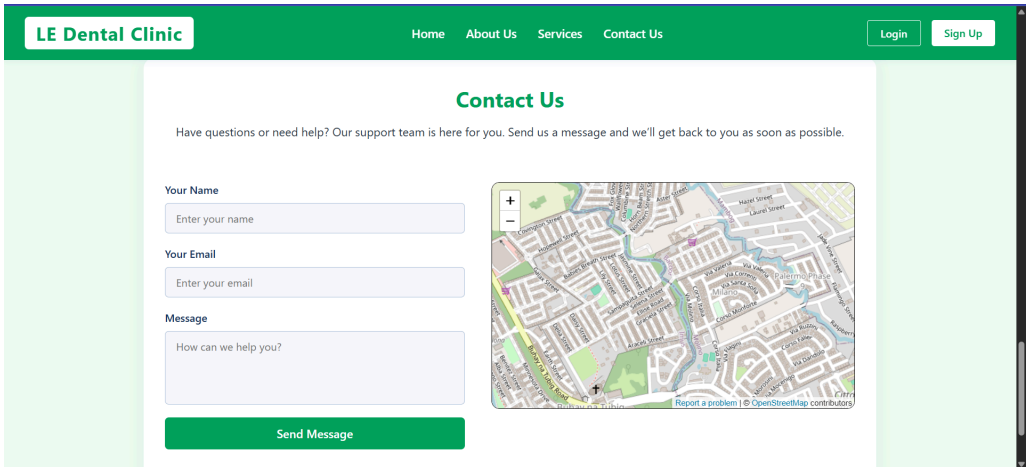
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Parts of the System

Patient Side

Landing Page



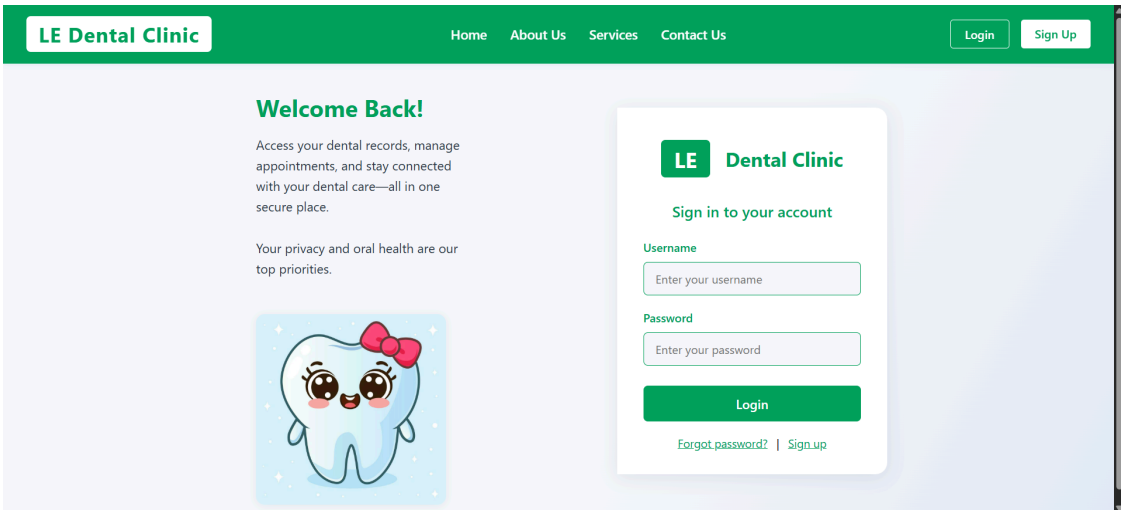


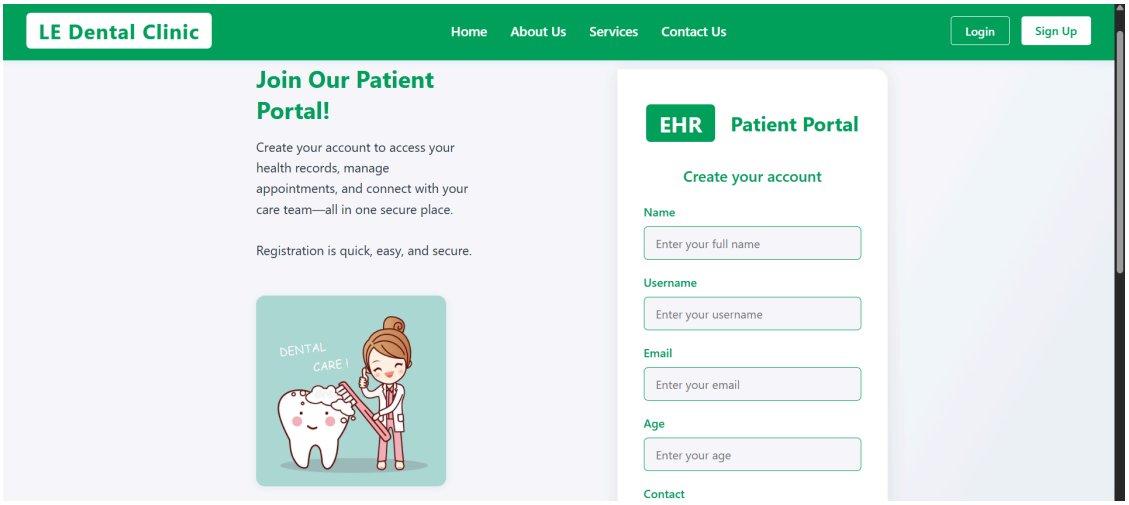
The landing page contains simple information regarding the proposed Healthcare Management System. In this page, its main functionalities, at least for the side of the Patient users, is stated and displayed to help the patient easily navigate the whole system. The page also allows you to email the clinic.

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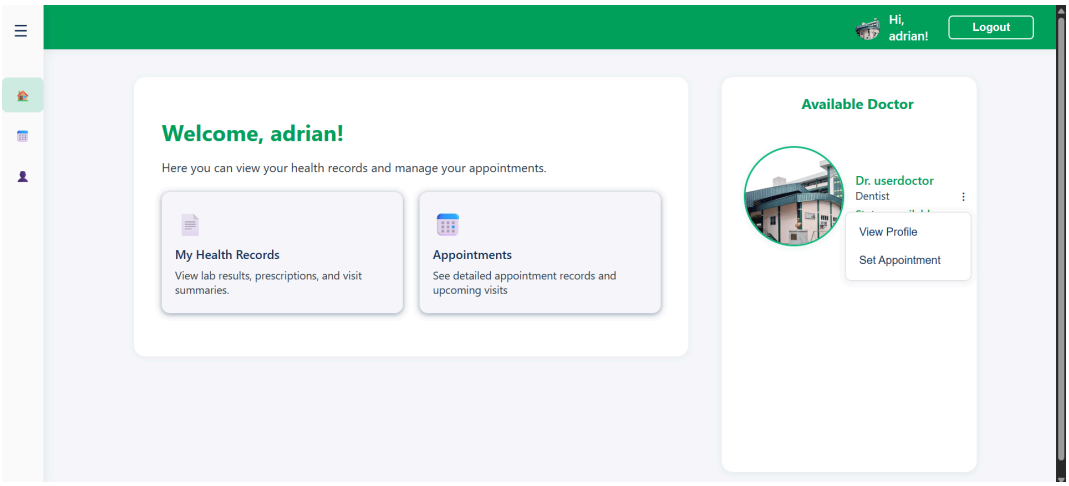
Login/Register Page

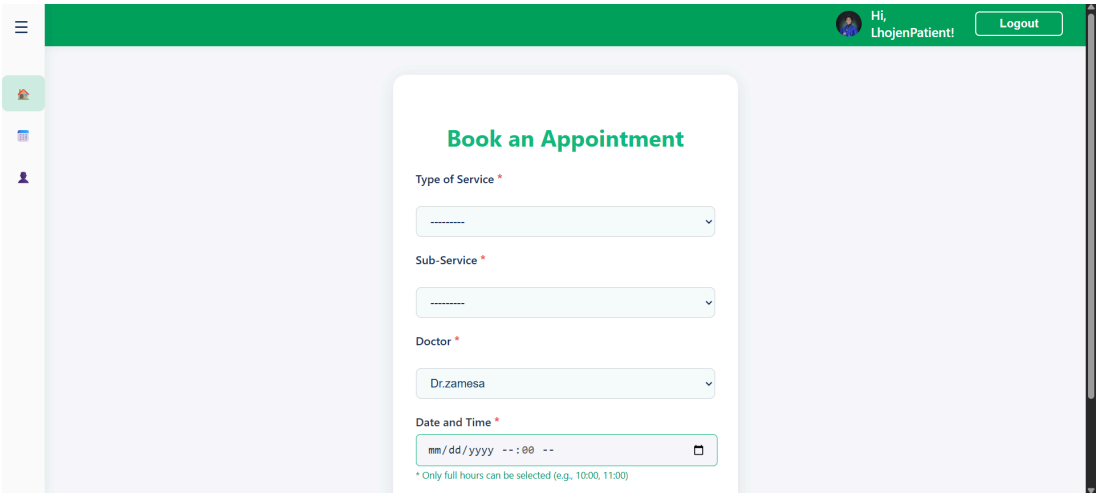




In the login/sign up page, this is where you can simply log in assuming the user/patient already has an account. However, if the user has no currently existing account within the system, the user can register or sign up using the sign up form, and login afterwards after the process has been done.

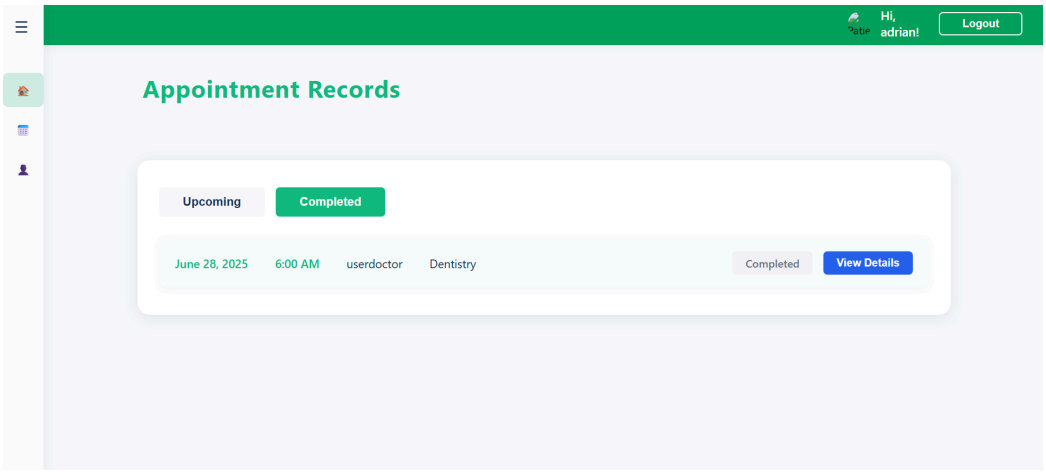
Patient Dashboard

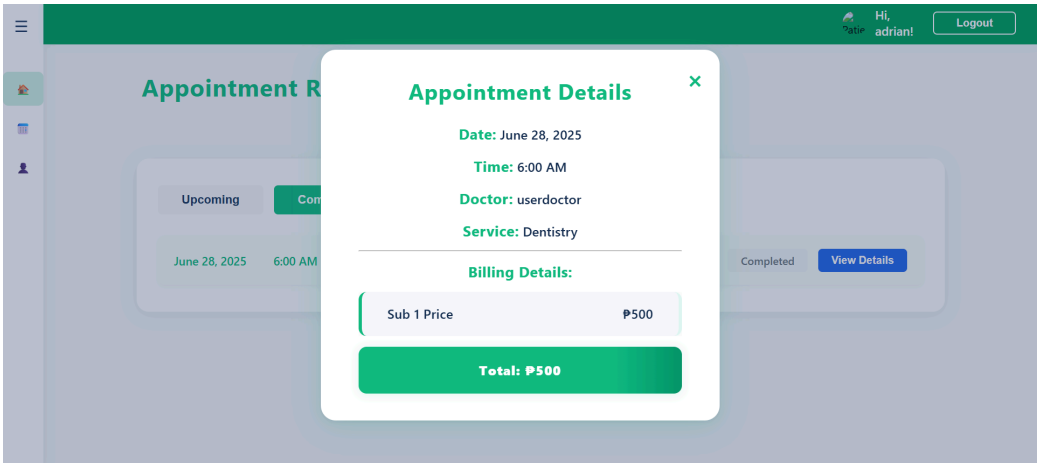




Within this page, the patient can view all of the listed doctors within the said facility. After clicking on the vertical ellipsis, they can set an appointment directly with the selected doctor. Once all the details in the appointment form have been filled up, the data will be recorded and sent to the doctors for its verdict.

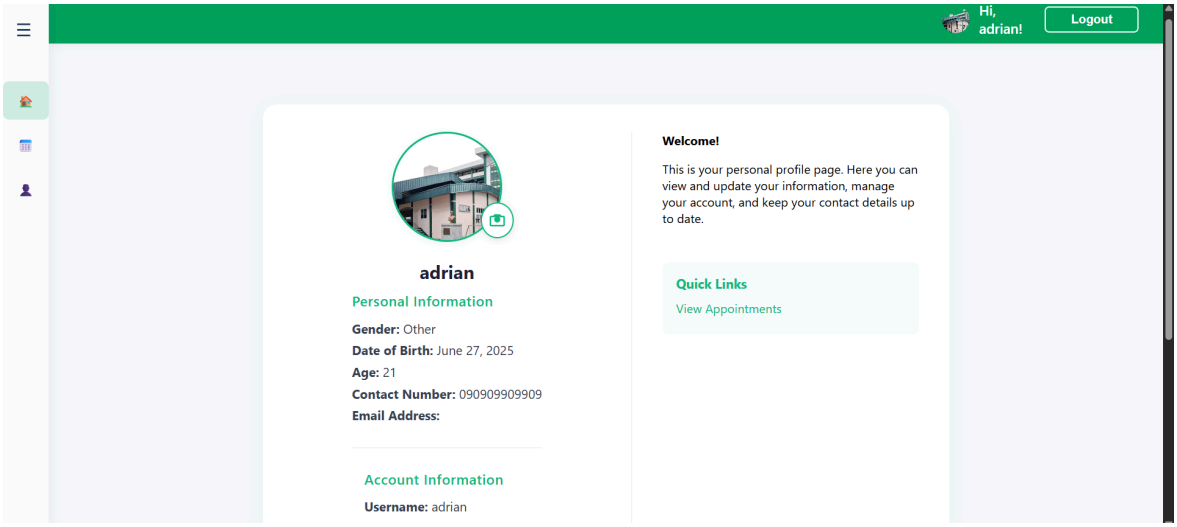
Appointment Records

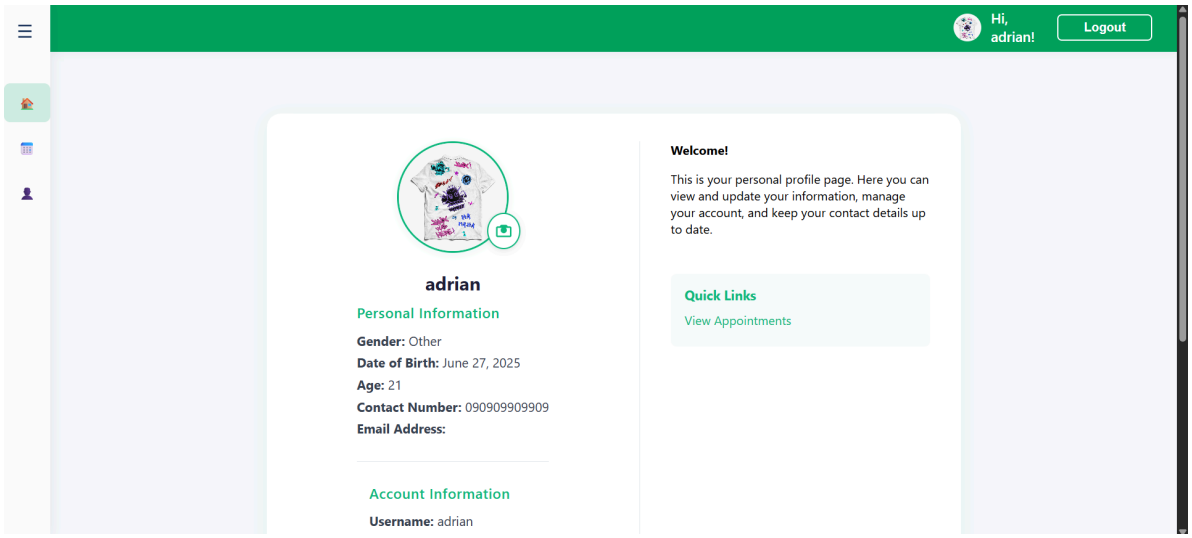




This form simply allows the patient to view all of their previous appointment schedules whether it's pending, confirmed, or completed. This page will also allow the patient to keep track about the billing information regarding the specific appointment schedule that was completed.

Patient Profile

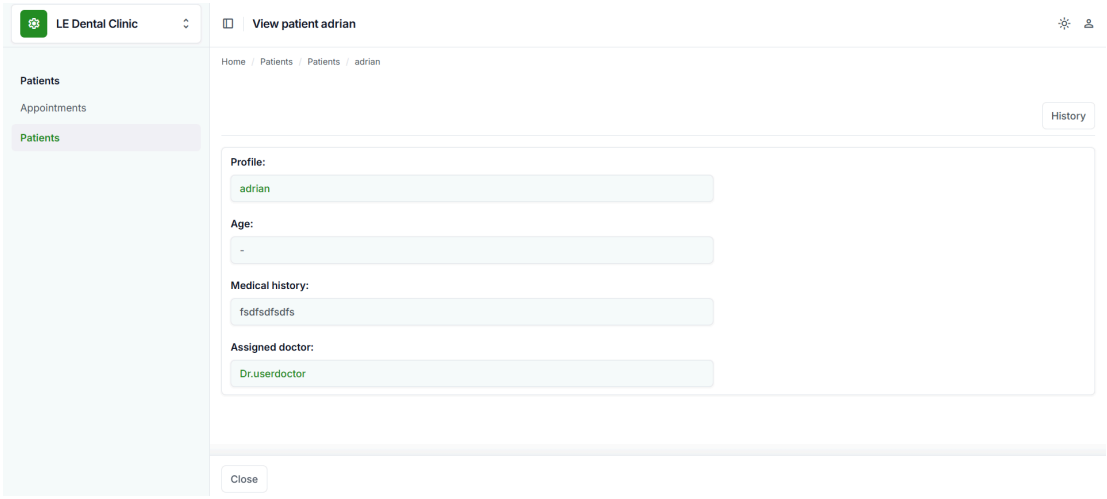


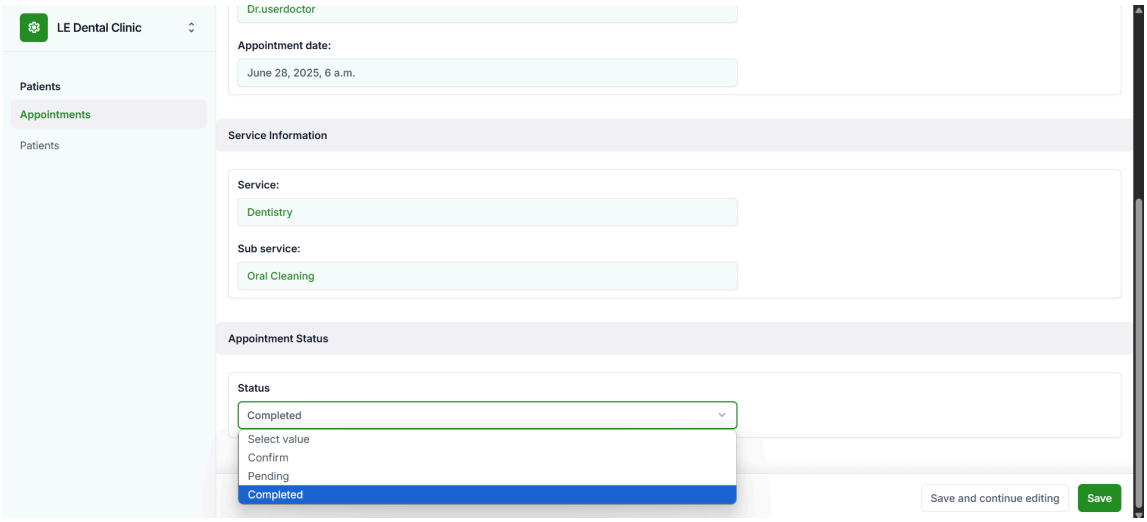


The Patient Profile page is the simplest of all the discussed pages. This page simply allows the patient to view the details of their accounts. It also allows the user to change their profile picture.

Doctor Side

Patients Administration



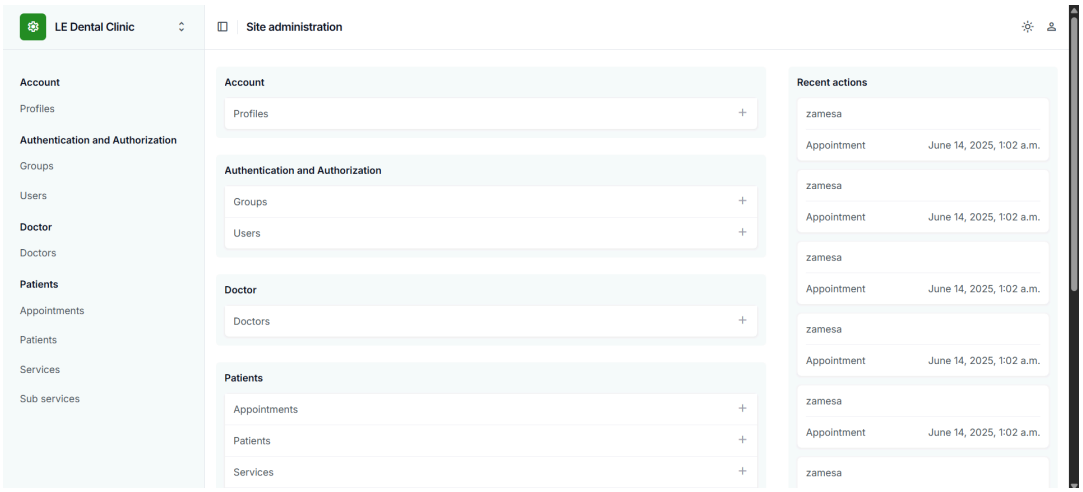


To put it simply, this page allows the doctor to change the verdict of the appointment made by the patient (complete, confirm, pending, no-show), and to view the details of the listed patients. The admin, while they can view all of these data, cannot change the recorded values. The only value they can change would be the verdict or the status of the appointment.

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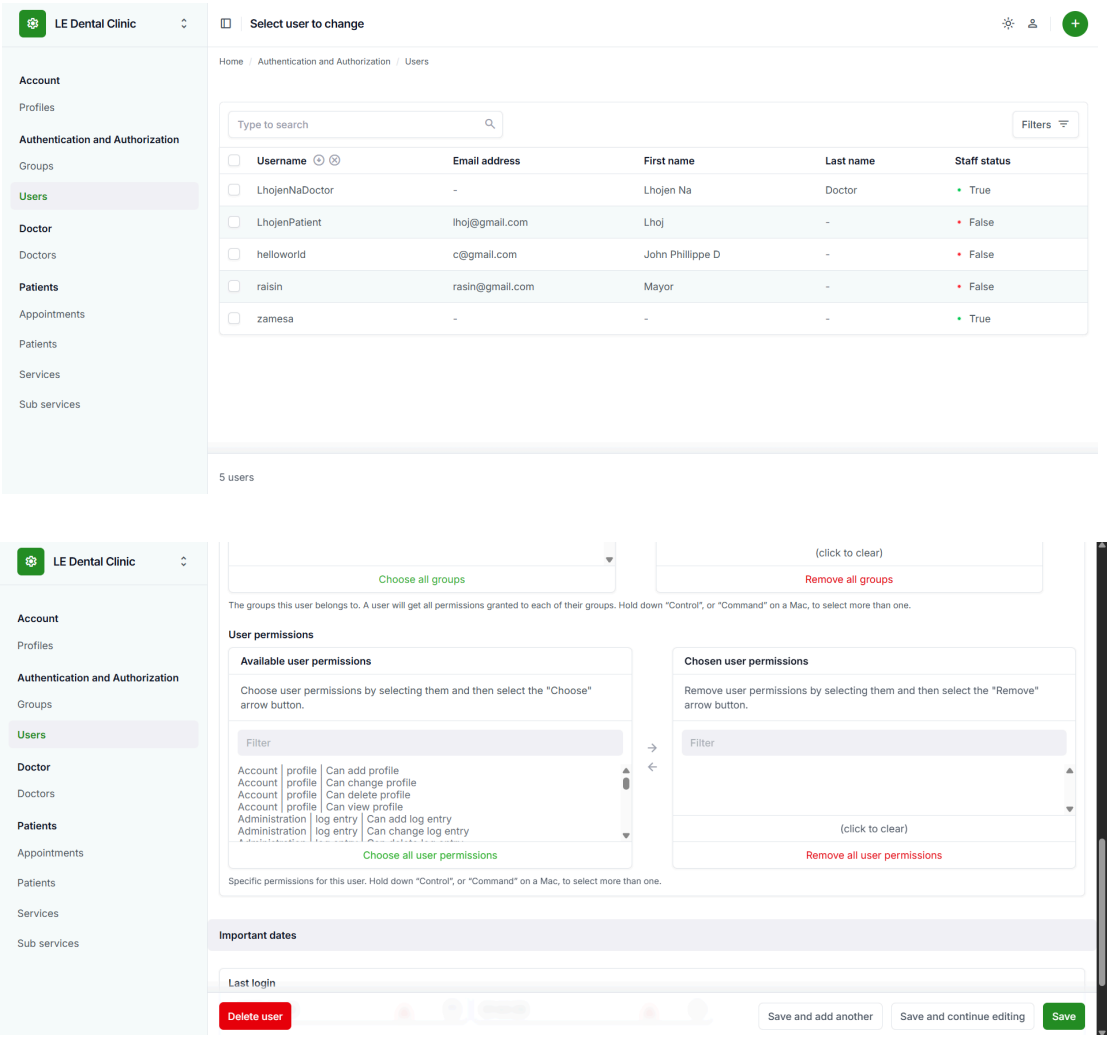
Admin Side

Site Administration



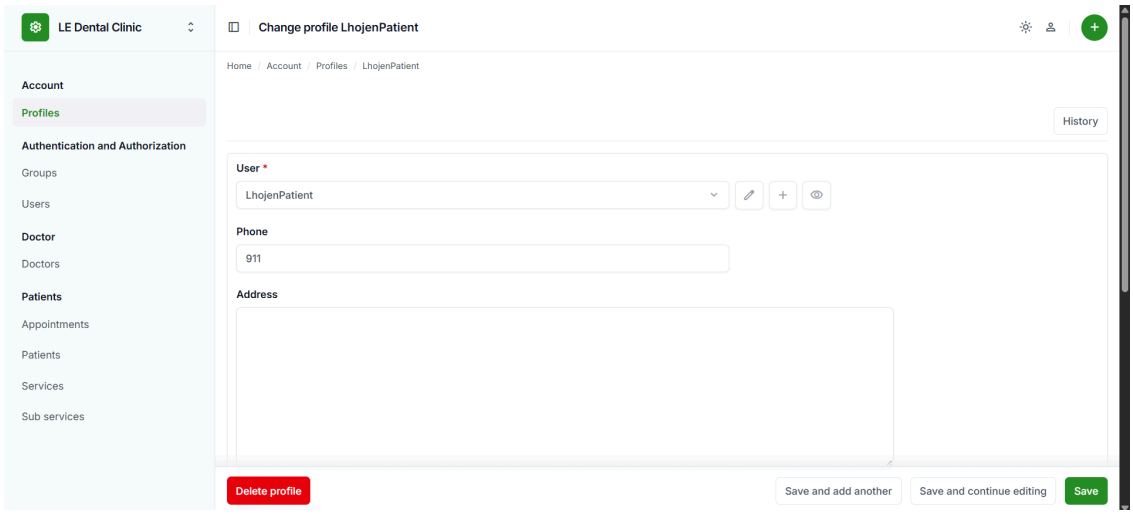
In the Site Administration, the Admin can manage different information within the system such as appointment records, user and profiles of both patients and doctors, the basic information of patients and doctors, as well as the management of services and sub services offered by the facility. To put it simply, this page allows the admin to manage information within the said system.

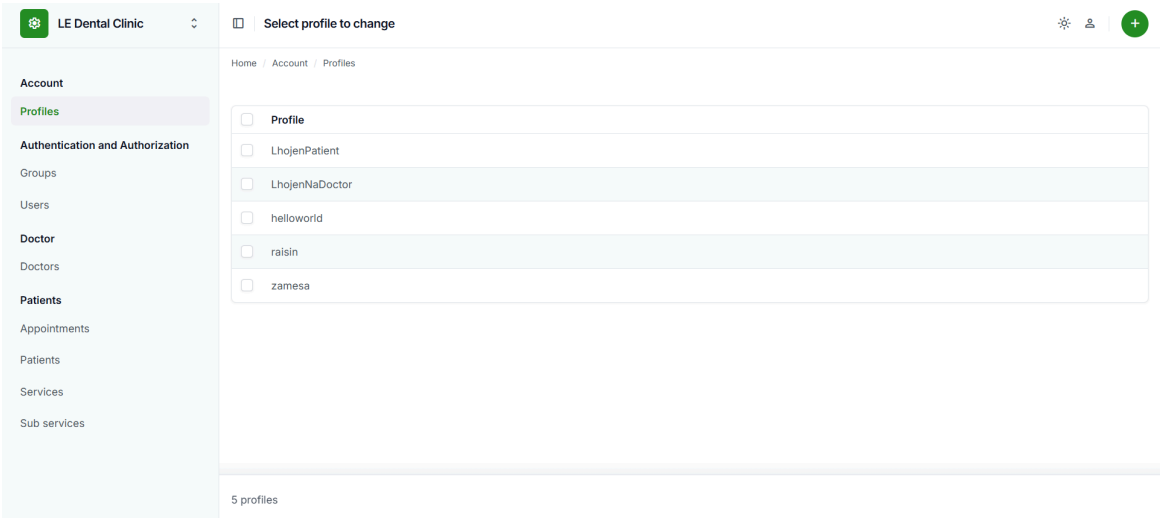
Authentication and Authorization



This page allows the admin to view all the users that are stored within the database. It also allows the admin to give users permissions, depending on the role they have on the system/facility. Also, keep in mind that every created user has no user type, and needs to be configured within the Profile settings and then registered within the Doctor/Patient records. Admins are also allowed to change the details and permissions if needed be, except for the password.

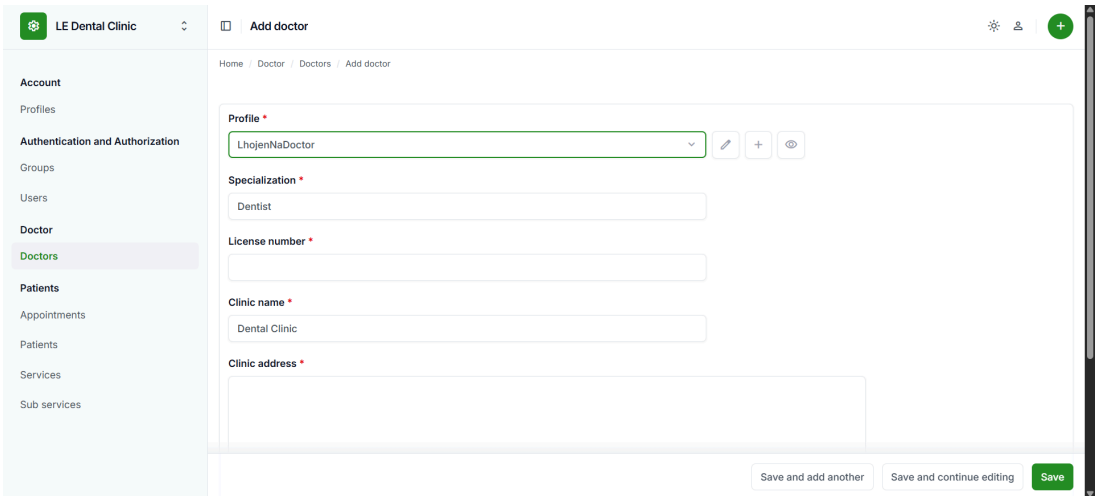
Account

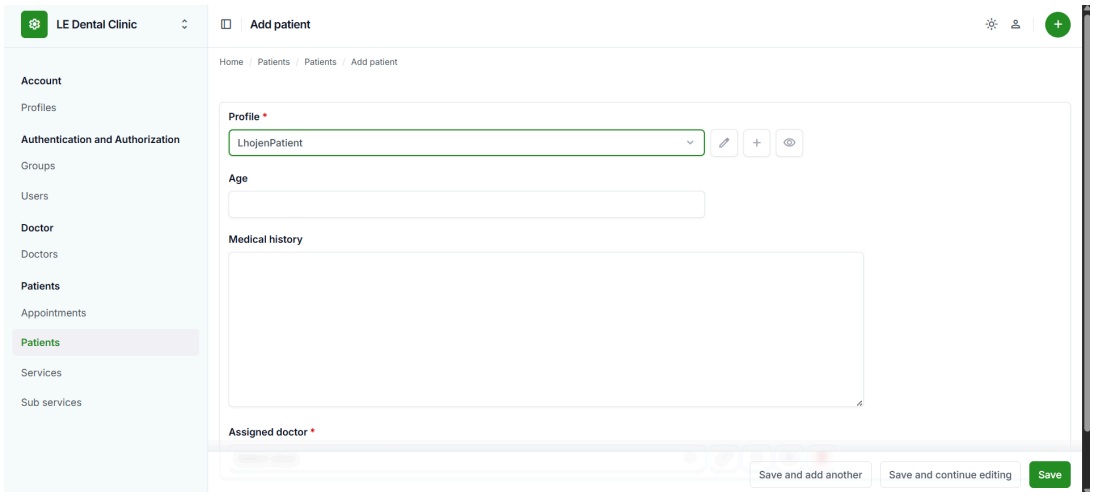




In this page, admins are allowed to create the profile of the listed users to verify them to be viable for the Patient and Doctor records. When a Patient creates an account using the sign up option, they would not be immediately registered within the Profiles and Patients record. They would need to be registered by the admins. Admins are also allowed to change the details of these profiles.

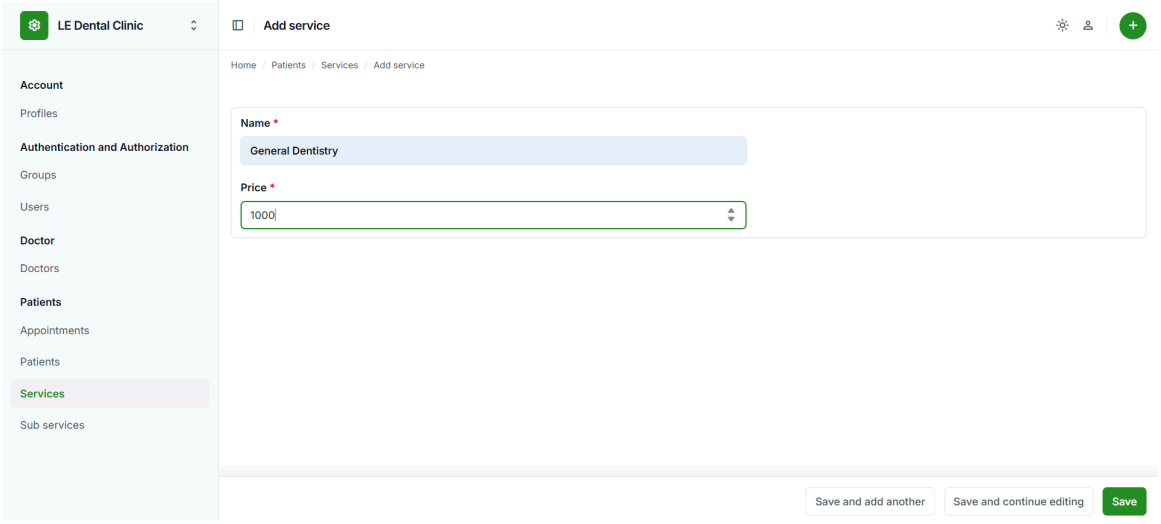
Patient and Doctor Records

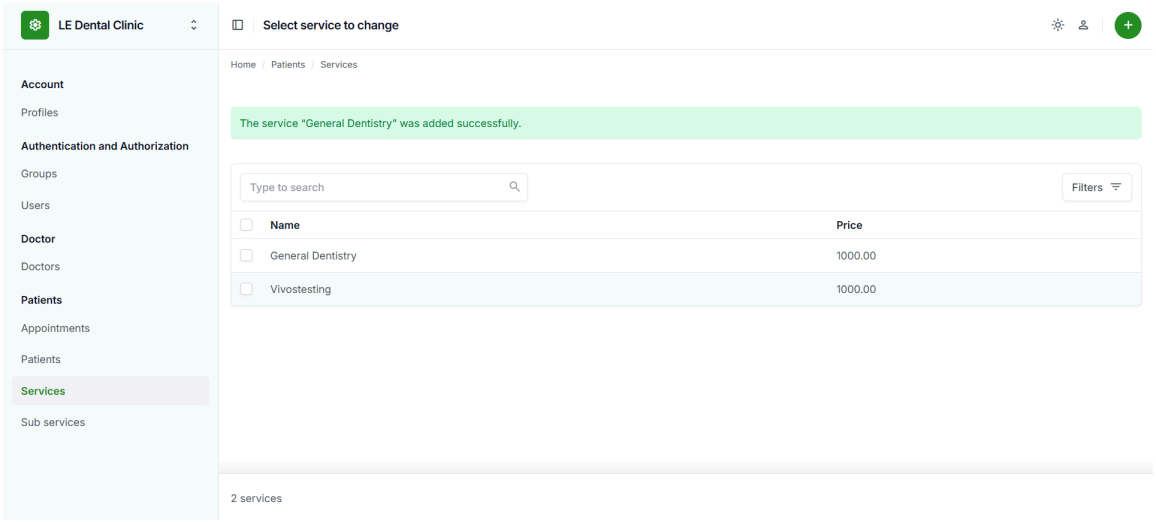




After registering the profile of the users, they can now be officially registered within the Patient and Doctor records. Aside from adding and deleting Patients and Doctors within the database, this page also allows admins to change details of both patients and doctors, with information such as medical history and clinic addresses being included.

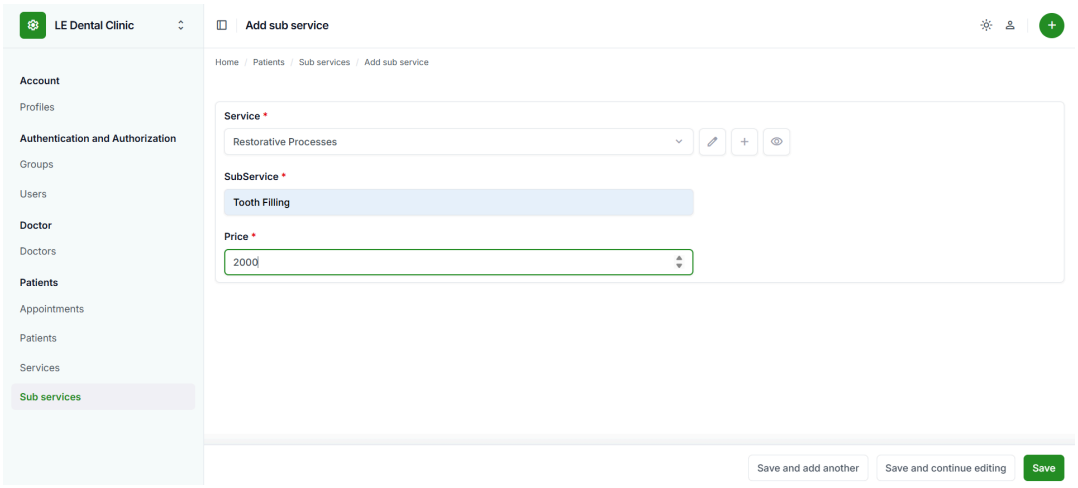
Services

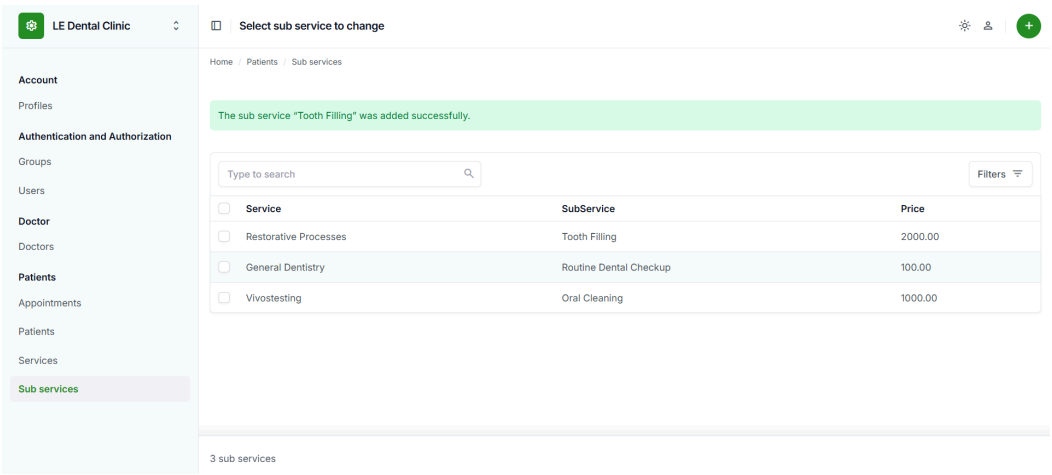




Admins are also allowed to add, change, delete, and view services that the facility will offer. Ifr need be, the admins can change the price and name of the services. Keep in mind that the admin needs to create a service first in order to create a sub service.

Sub Services





Once the service has been created, the admin can now create the sub services that are under a specific service. In addition, if need be, they can change the details of the specific sub service they want to update, as well as delete these sub services.